

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	4	9	18

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

let $\hat{y}_i = X \hat{\beta}$

Given,

$$X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}$$

$$Y = X\theta + \epsilon \Rightarrow \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{bmatrix}$$

$$\text{LSE of } \theta \quad Q(\theta_1, \theta_2) = \sum_{i=1}^3 \epsilon_i^2 = \sum_{i=1}^3 (y_i - x_i\theta)^2 = \sum_{i=1}^3 (y_i - X\theta)^2$$

$$= (Y - X\theta)^T (Y - X\theta)$$

$$= (Y^T - \theta^T X^T) (Y - X\theta)$$

$$= Y^T Y - \theta^T X^T Y - Y^T X \theta + \theta^T X^T X \theta$$

$$Q(\theta_1, \theta_2) = Y^T Y - 2\theta^T X^T Y + \theta^T X^T X \theta$$

w.k.t $\frac{\partial}{\partial n} [a^T n] = a$ and $\frac{\partial}{\partial n} [n^T a n] = (a + a^T) n$

derivative of $Q(\theta_1, \theta_2)$ w.r.t. θ .

$$-2X^T Y + 2X^T X \theta = 0$$

$$\hat{\theta} = (X^T X)^{-1} X^T Y$$

$$\hat{\theta} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} = (X^T X)^{-1} X^T y.$$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}$$

Row 1

$$1+1+4$$

$$1-2-2$$

$$1-2-2$$

$$1+4+1$$

$$= \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{(36-9)} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix}$$

$$= \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$(X^T X)^{-1} (X^T) = \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 3 & 0 & 3 \\ 3 & -3 & 0 \end{bmatrix}$$

$$(X^T X)^{-1} X^T = \frac{1}{3} \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\text{LSE of } \hat{\theta}_1 = \frac{y_1 + y_3}{3}$$

$$\text{LSE of } \hat{\theta}_2 = \frac{y_1 - y_2}{3}$$

$$\hat{\theta} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} \frac{y_1 + y_3}{3} \\ \frac{y_1 - y_2}{3} \end{bmatrix}$$

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

a

Given,

U_1 & U_2 are 2 independent variable,

$$X_1 = U_1, X_2 = U_2 \text{ and } X_3 = U_1 + U_2$$

$$\begin{aligned} \text{Var}(X_1) &= \text{Var}(U_1) \\ &= \frac{1}{12} \end{aligned}$$

$$E(U_1) = \frac{1}{2}$$

$$\begin{aligned} \text{Var}(U_1) &= \frac{1}{12} - \left(\frac{1}{2}\right)^2 \\ &= \frac{1}{12} \end{aligned}$$

$$\begin{aligned} \text{Var}(X_2) &= \text{Var}(U_2) \\ &= \frac{1}{12} \end{aligned}$$

$$\begin{aligned} \text{Var}(X_3) &= \text{Var}(U_1 + U_2) \quad \left\{ \begin{array}{l} \text{since } U_1, U_2 \text{ are} \\ \text{independent} \end{array} \right. \\ &= \text{Var}(U_1) + \text{Var}(U_2) + 2\text{Cov}(U_1, U_2) \\ &= \frac{1}{6} \quad \rightarrow \underline{0} \end{aligned}$$

$$\text{Cov}(X_1, X_2) = \text{Cov}(U_1, U_2)$$

$$= E(U_1 U_2) - E(U_1)E(U_2)$$

$$= \underline{0} \quad \rightarrow \text{independent} \Rightarrow E(U_1 U_2) = E(U_1)E(U_2)$$

$$\begin{aligned} \text{Cov}(X_1, X_3) &= \text{Cov}(U_1, U_1 + U_2) \\ &= \text{Cov}(U_1, U_1) + \text{Cov}(U_1, U_2) \\ &= \text{Var}(U_1) \\ &= \frac{1}{12} \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_2, X_3) &= \text{Cov}(U_2, U_2 + U_1) \\ &= \text{Cov}(U_2, U_2) + \text{Cov}(U_2, U_1) \\ &= \frac{1}{12} = \text{Var}(U_2) \end{aligned}$$

$$\text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)} \sqrt{\text{Var}(X_2)}} = 0$$

$$\text{Corr}(X_1, X_3) = \frac{\text{Cov}(X_1, X_3)}{\sqrt{\text{Var}(X_1)} \sqrt{\text{Var}(X_3)}} = \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \sqrt{\frac{1}{6}}} = \frac{1}{\sqrt{12}} = \frac{1}{\sqrt{2}}$$

$$\text{Corr}(X_2, X_3) = \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \sqrt{\frac{1}{6}}} = \frac{1}{\sqrt{2}}$$

03

$$P = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \frac{1}{12} & 0 & \frac{1}{12} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} \end{bmatrix}$$

b. $\text{Var}(a'X)$ should be maximum and $a'a=1$
 then $a'X$ called as first P.C

$$\text{Var}(a'X) = a' \Sigma a$$

w.k.t $a' \Sigma a$ is maximum at $a = e_1$

where e_1 is the eigen vector of Σ

$$K = \begin{bmatrix} \frac{1}{12} - \lambda & 0 & \frac{1}{12} \\ 0 & \frac{1}{12} - \lambda & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} - \lambda \end{bmatrix}$$

$$\det(K) = 0$$

$$\Rightarrow \left(\frac{1}{12} - \lambda\right) \left[\left(\frac{1}{12} - \lambda\right) \left(\frac{1}{6} - \lambda\right) - \frac{1}{144} \right] + \frac{1}{12} \left[-\frac{1}{12} \left(\frac{1}{12} - \lambda\right) \right] = 0$$

$$\Rightarrow \left(\frac{1}{12} - \lambda\right) \left[\left(\frac{1}{12} - \lambda\right) \left(\frac{1}{6} - \lambda\right) - \frac{1}{72} \right] = 0$$

$$\Rightarrow \left(\frac{1}{12} - \lambda\right) \left[-\frac{\lambda}{6} + \lambda^2 \right] = 0$$

$$\lambda_1 = \frac{1}{12}, \lambda_2 = 0, \lambda_3 = \frac{1}{4} \quad \max(\lambda_1, \lambda_2, \lambda_3) = \frac{1}{4}$$

$\therefore$ Hence Variance of 1st PC = $\frac{1}{4}$

3. Let the matrix of distance be

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{matrix}$$

- (a) (6 points) Cluster the four items using each of the following procedures:
- Complete linkage hierarchical procedure.
 - Average linkage hierarchical procedure.
- (b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

② Given,

initial distances

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{matrix}$$

i. Complete linkage hierarchical procedure.

$$d_{(uv)w} = \max \{ d_{u,w}, d_{v,w} \}$$

first $\Rightarrow$ least distance is (1,2)

$\Rightarrow$ club those (1,2)

remove the columns and rows of 1,2

add (12)

new matrix

$$\begin{matrix} & \begin{matrix} (12) & 3 & 4 \end{matrix} \\ \begin{matrix} (12) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 11 & 0 & 0 \\ 5 & 3.5 & 0 \end{bmatrix} \end{matrix}$$

$$d_{(12)3} = \max(d_{13}, d_{23})$$

$$d_{(12)4} = \max(d_{14}, d_{24})$$

$$d_{(2)3} = \max(11, 2)$$

$$d_{(12)(4)} = \max(5, 3)$$

now

least distance between them is 3 and 4.

we will remove 3 and 4 columns and rows and add (34) column and row

$$\begin{array}{c} (12) \\ (34) \end{array} \begin{array}{cc} (12) & (34) \\ \left[\begin{array}{cc} 0 & 0 \\ 11 & 0 \end{array} \right] \end{array} \quad \begin{array}{l} d_{(12)(34)} = \max [d_{12,3}, d_{12,4}] \\ d_{(12)(34)} = \max [11, 5] \\ = 11 \end{array}$$

last club remaining two

(1234) = final cluster.

ii.

Average linkage hierarchical procedure.

$$d_{(uv)(w)} = \frac{\sum_{i=1}^{N(uv)} \sum_{j=1}^{N(w)} d_{ij}}{N(uv) N(w)}$$

$N(uv)$ = no of nodes in Total uv cluster

$N(w)$ = no of nodes in w cluster.

first lowest $\Rightarrow$ (1, 2)
 $\hookrightarrow$ add (12) remove 1 & 2

$$\begin{array}{c} (12) \\ 3 \\ 4 \end{array} \begin{array}{ccc} 3 & 4 & \\ \left[\begin{array}{cc} 0 & \\ 6.5 & 0 \\ 4 & 3.5 \end{array} \right] \end{array} \quad \begin{array}{l} d_{(12)(3)} = \frac{d_{13} + d_{23}}{2(1)} \\ = \frac{11 + 2}{2} = \frac{13}{2} = 6.5 \\ d_{(12)(4)} = \frac{d_{14} + d_{24}}{2} = \frac{5 + 3}{2} = 4 \end{array}$$

Cont

lowest distance is (3,4)

add (3,4) remove 3,4

new distance

	(12)	(34)
(12)	0	8
(34)	5.25	0

$$d_{(12)(34)} = \frac{d_{13} + d_{14} + d_{23} + d_{24}}{2(2)}$$

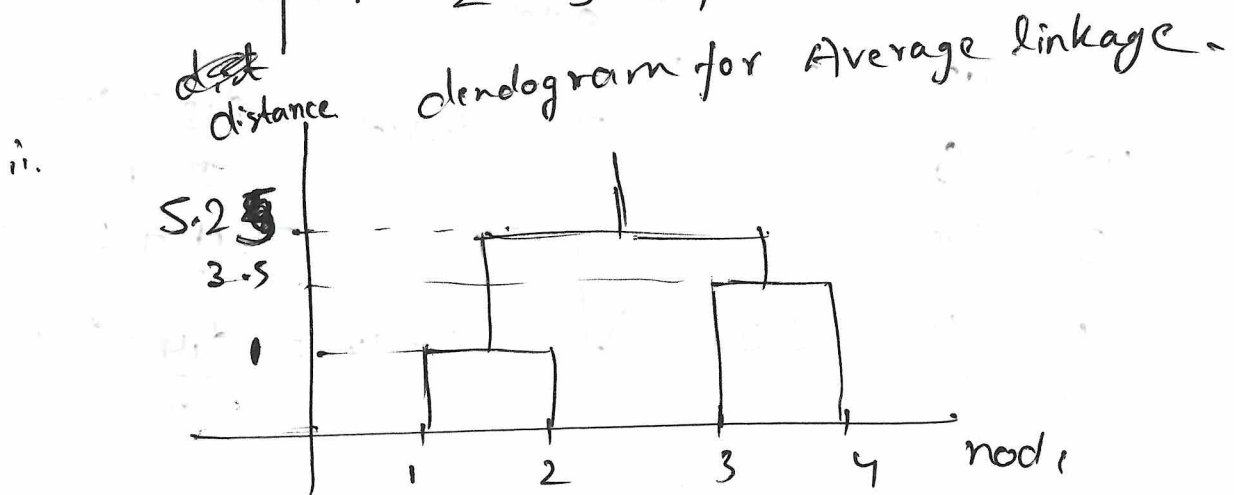
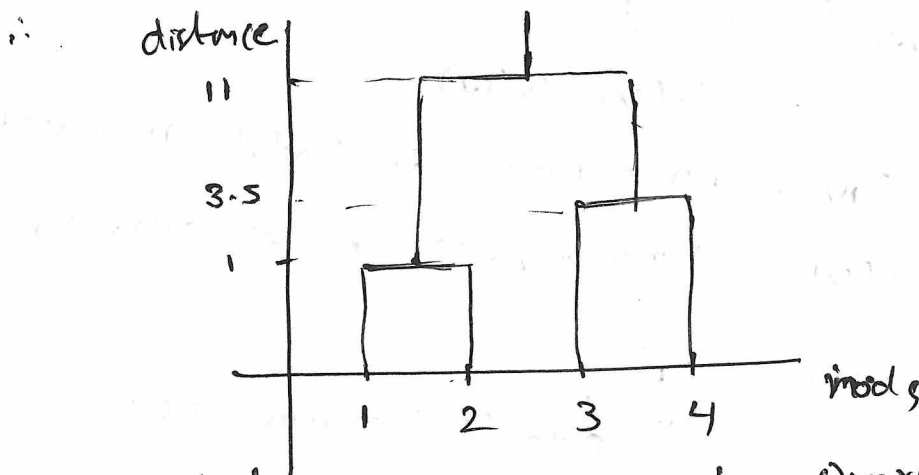
$$d_{(12)(34)} = \frac{11 + 5 + 2 + 3}{4}$$

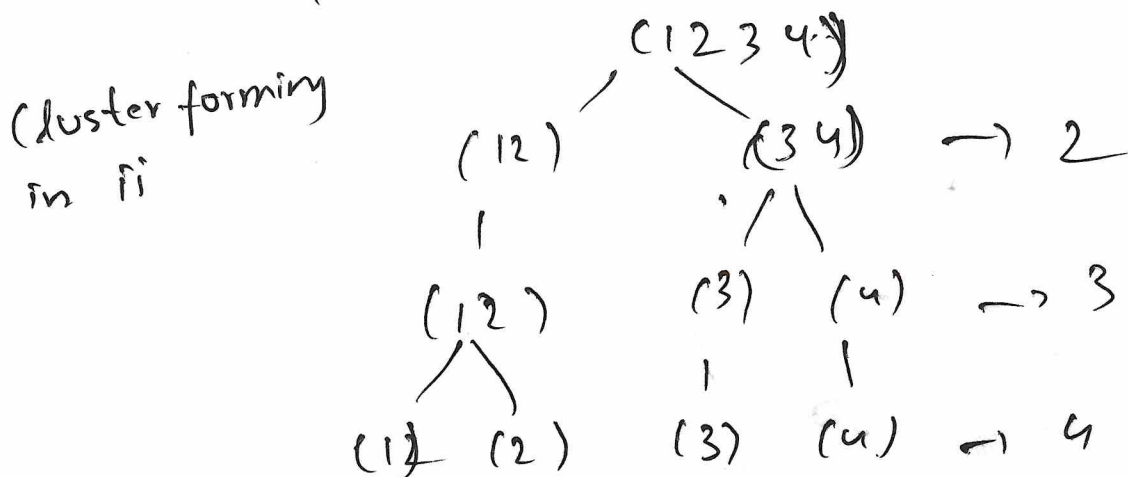
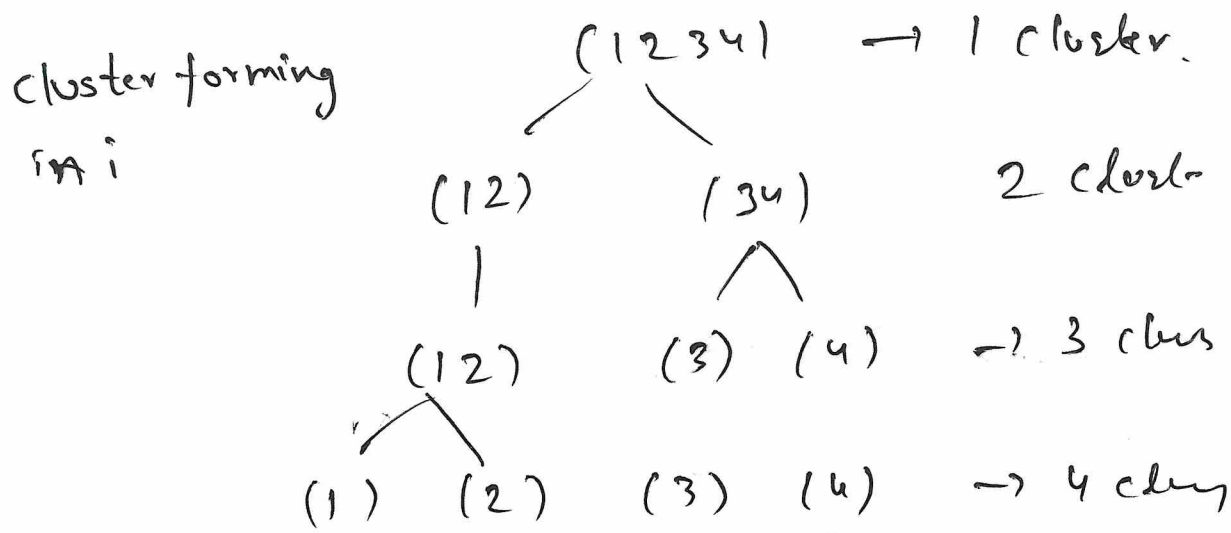
$$= \frac{16 + 5}{4} = \frac{21}{4} = 5.25$$

(06)

⇒ next club (12), (34) to have a single cluster (1234)

b. dendrogram for complete linkage.





→ both are forming cluster same, but usage of linkage is different.

in i) we can say that 1 & 2 are forming at a distance of 1.

and (3,4) are forming at a distance of 3.5.

here both 3 and 4 are farther from cluster (1,2) i.e. $\{ \text{dist}(1,3) \text{ and } \text{dist}(2,3) \}$ is more than distance of 3 and 4; so, 3 form cluster with 4

ii) In average linkage, $\frac{\text{dist}(1,3) + \text{dist}(2,3)}{2} > \text{dist}(3,4)$

03 so, (3,4) form cluster.

